

MATH 460 – Solutions Journal

David Gonzalez
gonz9032@bears.unco.edu

September 9, 2026

Notes: *Solutions Journal for Math 460, Introduction to complex analysis. All problems worked on and revised will be kept here.*

Chapter 0: The Real Line

0.1: Functions as Transformations

Task 1) This is the first task!! **Notes (Chapter 0, Task 1):** *These are notes.*

BELOW IS THE TEMPLATE WITH EXAMPLES

Problem 1

This is the first problem.

This is the second paragraph of the first problem. This equation $c^2 = a^2 + b^2 - 2ab \cos(\theta_C)$ is an example of in-line math.

And this is an example of display mode math:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

This is an example of the align* environment, which can be helpful for mathematical expressions with multiple steps:

$$\begin{aligned} r &= 1/2 + 1/3 + 1/4 + 1/5 + 1/6 + 1/7 + 1/8 + \dots + 1/n \\ &> 1/2 + 1/4 + 1/4 + 1/8 + 1/8 + 1/8 + 1/8 + \dots + 1/n \\ &= \sum_{i=1}^{\log_2 n} \frac{2^{i-1}}{2^i} \\ &= \frac{1}{2} \log_2(n) \\ &= \Omega(\log n) \end{aligned}$$

This is an example of matrices in L^AT_EX:

$$\begin{aligned} \sigma_x &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ \sigma_y &= \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ \sigma_z &= \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{aligned}$$

This concludes the first problem and the mathematical examples. The following questions demonstrate how to use the template for multipart problems.

Problem 2

- a) This is the second problem, first question.
- b) This is the second problem, second question.
- c) This is the second problem, third question.

Problem 3

This is an introduction to the the third problem.

- a) This is the third problem, first question.

b) This is the third problem, second question.

c) This is the third problem, third question.

d) This is the third problem, fourth question.

Problem 4

And this is the final one.

And this is the final one.

And this is the final one.